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Catalytic Asymmetric Synthesis of Inherently Chiral Eight-Membered O-Heterocycles through Cross-[4+4] Cycloaddition of Quinone Methides

Da Xu, Guojie Zhou, Bangli Liu, Shiqi Jia, Yidong Liu, and Hailong Yan**

Abstract: Inherently chiral eight-membered rings embedded in tetraphenylene derivatives and hetero-analogues exhibit unique properties and allow diverse applications. A conceptually viable and straightforward approach to these frameworks is [4+4] cycloaddition, which still remains elusive. Herein, we describe the stereoselective cross-[4+4] cycloaddition of quinone methides (QMs), leading to the formation of oxa-analogues of tetraphenylene with exceptional chemo-, diastereo-, and enantioselectivity. The structures of these novel rigid eight-membered O-heterocycles were explored by single-crystal X-ray diffraction, and their stereochemical stability was elaborated through both density functional theory (DFT) calculations and thermal racemization experiments. The developed methodology exhibited broad substrate scope and the resulting cross-[4+4] cycloadducts could be readily transformed into valuable chiral building blocks. Our findings expand the library of inherently chiral medium-sized rings and also contribute to the advancement of asymmetric cross-[4+4] cycloadditions of quinone methides.

Introduction

Chirality is an intrinsic property of molecular systems. The design of new chiral molecular architectures and stereochemical control are significant for asymmetric synthesis.[1] Tetraphenylene with a rigid central eight-membered ring is formed by *ortho*-annulation of four benzenes.[2–5] The entire molecule with suitable substituents can create inherent chirality,[6–16] which is distinct from the prevailing stereogenic elements such as central, axial, planar, and helical elements. Due to their unique chiral properties, tetraphenylene derivatives are used as functional molecules and have attracted considerable attention.[17–19] However, only Shibata, Yang, and Zhu reported sporadic asymmetric synthetic approaches for the construction of chiral tetraphenylene derivatives and their aza-analogues (Figure 1A).[20–24] Notably, the challenge escalates when oxa-analogues are used as target molecules. The incorporation of oxygen into the central eight-membered ring increases the conformational flexibility, so the inherently chiral eight-membered O-heterocycles are elusive. Hence, it is significant to enhance its inversion barrier by the structural modification of oxa-analogue and develop the corresponding asymmetric catalytic method of enantioenriched oxa-analogues of tetraphenylene.

The [4+4] cycloaddition is conceptually recognized as a straightforward and atom-economical approach towards the eight-membered rings,[25,26] and has been successfully used to forge chiral cyclooctadiene products by Dieck, Cramer, and Jørgensen.[27–29] Nevertheless, the asymmetric synthesis of eight-membered O-heterocycles through thermal [4+4] cycloaddition was seldom reported due to the lack of synthetic strategies and suitable cycloaddends. *ortho*-Quinone methides (*o*-QMs) are highly reactive oxygenated four-atom synthons in synthetic chemistry, medicinal chemistry, and material chemistry.[30–34] The strong tendency of rearomatization makes them strong electrophiles and electron-poor heterodienes and valuable catalytic asymmetric reactions involving *o*-QMs or its analogues have been realized in recent years (Figure 1B).[35–53] Through the judicious choice of reagents, the asymmetric [4+3], [4+2], and [4+1] cycloadditions of *o*-QMs were well established.[54–67] The [4+4] cycloadditions of *o*-QMs are still underdeveloped, especially the cross-[4+4] cycloaddition of *o*-QMs (Figure 1B).[68,69] Except the control of the rapid reactivity of *o*-QMs, the above challenge may also arise from the unfavorable entropic and transannular interactions in the generation of medium-sized rings.[70–87]

Our group has been devoted to the development of asymmetric reactions by manipulating vinylidene *ortho*-quinone methides (VQMs), an *o*-QM analogue with an allene motif. With suitable substituent patterns, VQMs could retain the chiral feature and act as axially chiral intermediates in asymmetric reactions.[88–107] We aim to synthesize inherently chiral oxa-analogues of tetraphenylene through a conceptually straightforward and atom-economical cross-[4+4] cycloaddition of VQM and typical *o*-QM.

[*] D. Xu, G. Zhou, B. Liu, Y. Liu, Prof. Dr. H. Yan
Chongqing Key Laboratory of Natural Product Synthesis and Drug Research, Chemical Biology Research Center, School of Pharmaceutical Sciences, Chongqing University
Chongqing 401331 (P. R. China)
E-mail: yhl198151@cqu.edu.cn
yidong@cqu.edu.cn

S. Jia
Green Catalysis Center, and College of Chemistry, Zhengzhou University
100 Science Avenue, Zhengzhou 450001, Henan (P. R. China)

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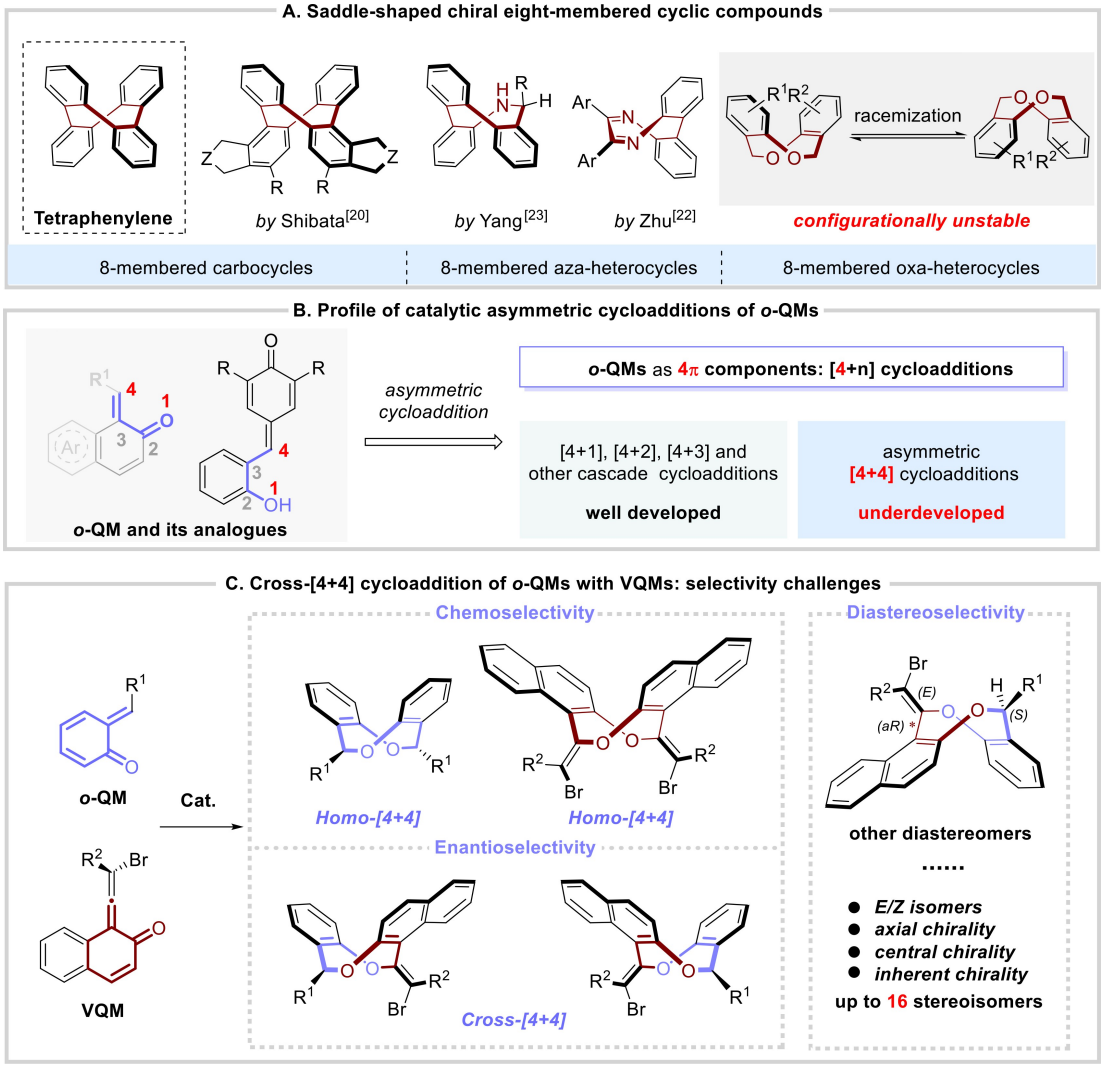


Figure 1. Catalytic enantioselective synthesis of saddle-shaped chiral eight-membered cyclic scaffolds by [4+4] cycloaddition.

However, due to the highly reactive nature of these polyene species, *o*-QMs and VQMs could generate various products due to complicated chemoselectivity (Figure 1C), such as [4+4]-isomers derived from competitive cross- and homo-cycloaddition and other polymerization products. Moreover, the thermal stepwise [4+4] cycloaddition could result in a large number of possible stereoisomers (Figure 1C). Except for selectivity, the stereochemical stability of the target saddle-shaped eight-membered O-heterocycles needs to be further verified.

Results and Discussion

Based on the above consideration, the preliminary computation was performed with *o*-QM **2a** and VQM **I** as model substrates to elucidate the possible chemically distinct products (Figure 2). The relatively stable and separable *o*-QM **2a** could efficiently compress the self-polymerization. VQM **I** could be generated *in situ* from the precursor 2-alkynylnaphthol **1a** in the presence of a Brønsted base catalyst and bromine source. The computation results of the cross-[4+4] cycloaddition of *o*-QM **2a** and VQM **I** revealed a stepwise process and the predicted barrier for the transition state **TS1** was 24.9 kcal/mol. The transition state **TS2** and **TS3** for the homo-[4+4] cycloaddition of VQM **I** or *o*-QM **2a** respectively resulted in significantly higher energy barriers, indicating that the cross-[4+4] cycloaddition of *o*-QM **2a** and VQM **I** yielded good chemoselectivity. Moreover, the calculation results of the pathways related to diastereoisomers indicated higher energy barriers (**TS4–6**) than **TS1**, thus leading to the predicted good diastereoselectivity.

Based on the initial calculation results, we examined the reaction between VQM **I** and *o*-QM **2a**. The methodology was a two-step one-pot protocol, in which the VQM **I** generation and cross-[4+4] cycloaddition were separately performed. Thus, the enantioselectivity of VQM **I** formation could be improved by tuning reaction conditions and the stereocontrol of the subsequent cross-[4+4] cycloaddition

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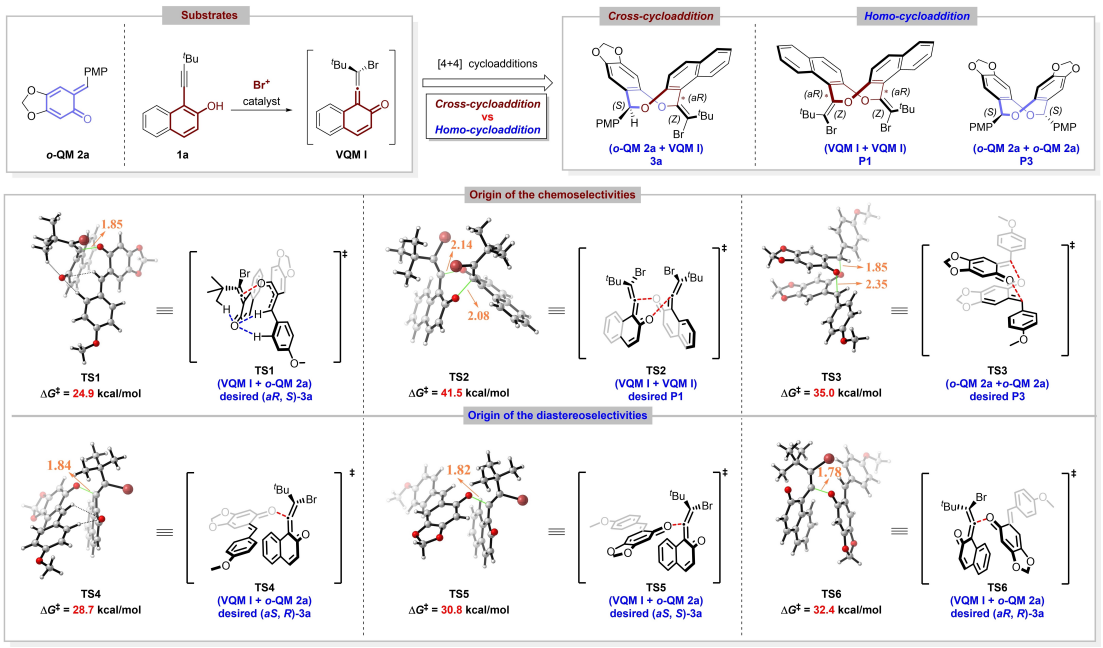
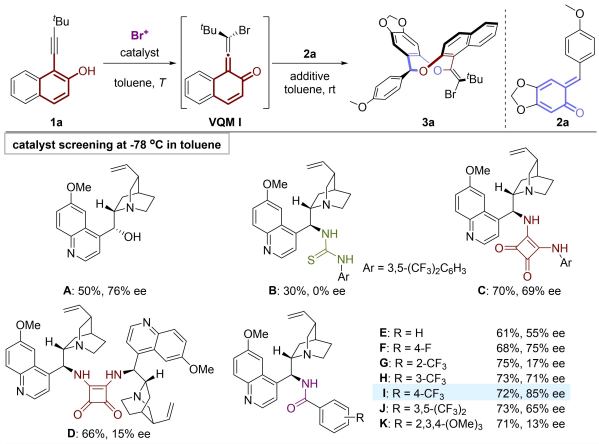


Figure 2. Understanding of the chemo- and diastereoselectivities through DFT calculations: optimized transition states corresponding to [4+4] cycloadditions of VQM **I** or *o*-QM **2a**. The bond lengths are in angstroms.

relied on the chirality conversion. Delightedly, the reaction between VQM precursor **1a** and NBS in toluene at −78 °C in the presence of quinine **A** delivered axially chiral VQM **I** intermediate and subsequent reaction quenching and addition of *o*-QM **2a** at room temperature afforded the cross-[4+4] cycloadduct **3a** in 50 % yield and 76 % ee with excellent diastereoselectivity (>20:1). The observed chemo- and diastereoselectivity of the cross-[4+4] cycloaddition was consistent with theoretical calculation results. To further improve the enantioselectivity of the whole process, a range of quinine derivatives **B**–**K** were examined (Table 1 and Supporting Information). Ultimately, catalyst **I** provided the highest selectivity for the construction of **3a**. A survey of different solvents revealed that toluene still had superior enantioselectivity (see Supporting Information). Increasing the reaction temperature in the first step did not improve the yield significantly, but the enantioselectivity was significantly changed (Table 1, entries 1–5). The enantioselectivity was better at −50 °C. At the concentration of 0.0125 M, **3a** was obtained in 91 % ee (Table 1, entry 7). Adding 1,4-diazabicyclo [2.2.2] octane (DABCO) as an additive in the second step further increased the yield to 85 % with the similar ee value (Table 1, entry 8), whereas 1,8-diazabicyclo [5.4.0] undec-7-ene (DBU) totally suppress the reaction (Table 1, entry 9).

With the optimal reaction conditions in hand (Table 1, entry 8), the generality of the cross-[4+4] cycloaddition was examined (Table 2). Firstly, the substituents on the phenyl ring at the exocyclic carbon of the *o*-QMs were evaluated. The phenyl ring bearing *para*-electron-donating groups, such as ethoxy, benzyloxy, and methylthio, exhibited good tolerance (**3b**–**d**). The phenyl ring featuring 3,4-dimethoxy and 3,4-methylenedioxy reacted smoothly, yielding the

Table 1: Optimization of the reaction conditions.[a]



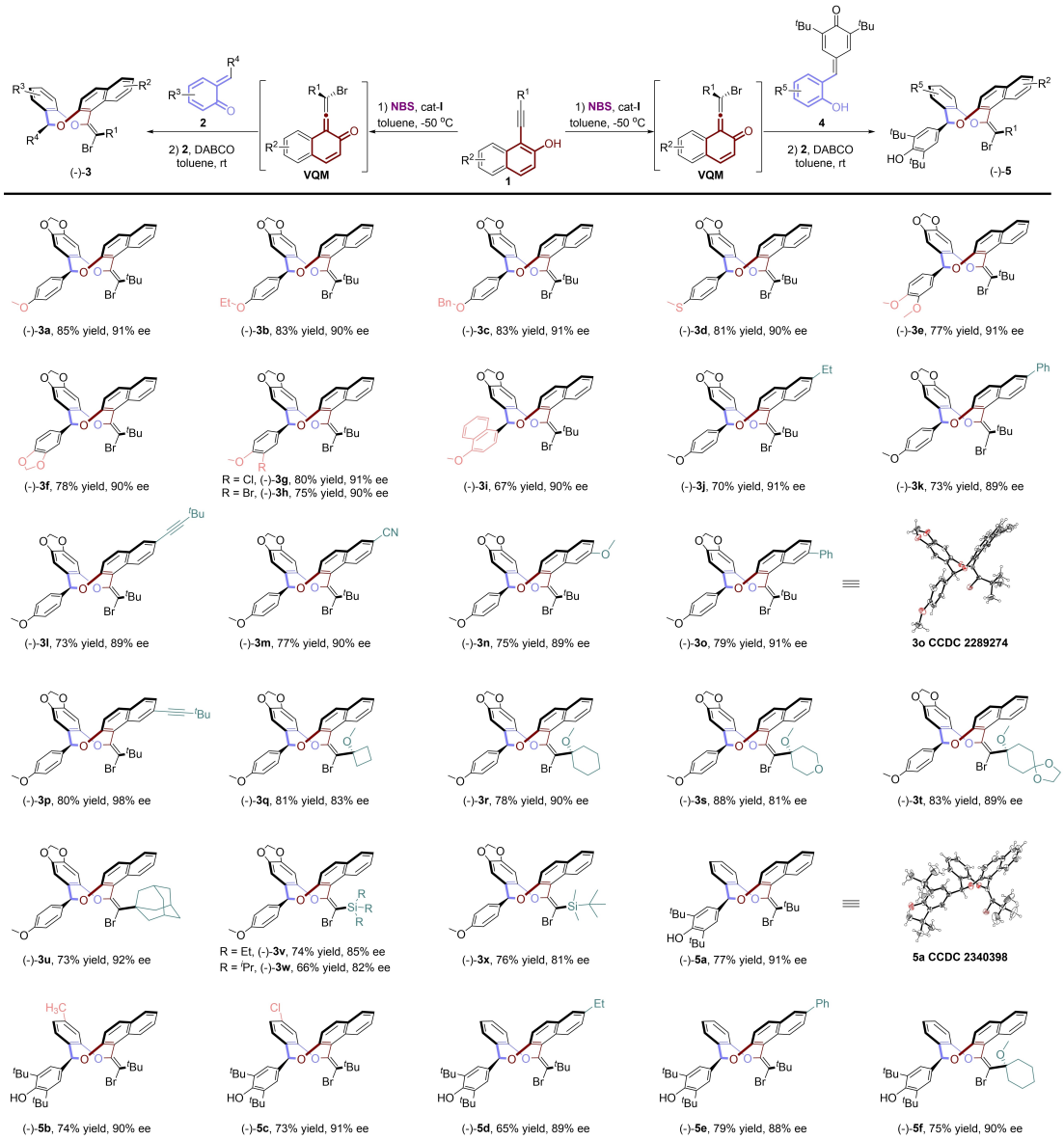
Entry	Catalyst	*T* (°C)	Additive	Yield[b] (%)	ee[c] (%)
1	I	0	-	75	45
2	I	-20	-	73	66
3	I	-40	-	73	87
4	I	-60	-	71	87
5	I	-50	-	72	88
6[d]	I	-50	-	72	90
7[e]	I	-50	-	72	91
8[e,f]	I	-50	DABCO	85	91
9[e,g]	I	-50	DBU	trace	-

[a] Reaction conditions: **1a** (0.1 mmol), catalyst (10 mol %) and NBS (0.12 mmol) in toluene (0.05 M) for 35 min. Quenched with H_2O, **2a** (0.12 mmol) was poured into the organic phrase under stirring conditions at rt for 72 h. [b] Yield of the isolated product. [c] Enantiomeric excess (ee) determined by HPLC. Diastereomeric ratios (d.r.) >20:1 in all cases. [d] Toluene (0.025 M). [e] Toluene (0.0125 M). [f] DABCO (0.02 mmol). [g] DBU (0.02 mmol).

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Table 2: Substrate scope.[a]



[a] Reaction conditions: **1** (0.1 mmol), cat–**I** (10 mol%) and NBS (0.12 mmol) in toluene (0.0125 M) at −50 °C for 35 min. Quenched with H_2O, **2** or **4** (0.12 mmol) and DABCO (0.02 mmol) were poured into the organic phase and stirred at rt for 72 h. [b] Yield of the isolated product. [c] Diastereomeric ratios (d.r.) > 20:1 in all cases. [d] The ee values were determined by HPLC analysis on a chiral stationary phase.

corresponding cycloadducts **3e** and **3f** in high yield with excellent enantioselectivity. The introduction of 3-chlorine or 3-bromine alongside 4-methoxy on the phenyl ring did not significantly affect reactivity or selectivity (**3g–h**). Changing the phenyl to methoxyl-substituted naphthyl was also compatible (**3i**). Next, we investigated the substituent R^2 of alkynylnaphthols. Ethyl, phenyl, aliphatic alkyne, cyano, and methoxyl groups at 6- or 7-position of the naphthol ring resulted in high yields with high to excellent enantioselectivity (**3j–p**). Furthermore, the influences of the substituents on the allene motif of VQMs were tested. The methoxy-substituted cyclobutyl, cyclohexyl or its variants, 1-adamantyl, and silane substituents, could uniformly afford the cross-[4+4] cycloadducts (**3q–x**). The structure and absolute configuration of the cross-[4+4] cycloadduct **3o** were determined through X-ray diffraction analysis.

In addition, *ortho*-hydroxyphenyl substituted *para*-quinone methides (*p*-QMs) **4** were investigated. Under the standard reaction conditions, *ortho*-hydroxyphenyl substituted *p*-QM **4a** participated smoothly in the cross-[4+4] cycloaddition, thus endowing cycloadduct **5a** with high efficiency and excellent enantioselectivity. Both methyl and chlorine in the *para* position of the aryl ring of **4** were well tolerated (**5b–c**). VQM precursors with ethyl or phenyl at the naphthol or methoxy-substituted cyclobutyl at alkyne moiety were also compatible to afford desired products (**5d–f**). The structure and relative configuration of product **5a** were determined using single-crystal X-ray diffraction. The

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absolute configuration of **5e** was confirmed by comparing the experimental electronic circular dichroism (ECD) spectrum with the results of time-dependent density functional theory (TD-DFT) calculations. The stereochemistry of all the other products was assigned by analogy.

The crystallographic analysis revealed that compound **3o** possessed a bridged axially chiral styrene with an eight-membered linkage. The entire molecule featured a saddle-shaped geometry and the central eight-membered O-heterocycle was a boat conformation (Figure 3A). The stereochemical stability of **3a** was then probed by thermal racemization experiments and the enantiomeric and diastereomeric ratios in toluene respectively at 60 °C and 90 °C for 8 h showed no significant change (see Supporting Information). To further investigate the stability of eight-membered O-heterocycles conformers, we calculated the inversion barriers for compounds **3a** and their analogs (**3–I** with central chirality removed, **3–III** with olefin axial chirality removed, and **3–V** with both central and axial chirality removed). The computational results indicate that the synthesized oxa-eight-membered ring exhibits significant rigidity, with an interconversion energy barrier of 48.2 kcal/mol. This rigidity is primarily attributed to the bulky alkenes. while the stereogenic carbon center of compound **3a** has only a marginal influence on the interconversion barrier (Figures 3C–E).

To elucidate the synthetic potential of these cycloadducts, we performed various transformations of compounds **3a** and **5a** (Figure 4). The treatment of **3a** with aluminum trichloride led to the cleavage of one ether group with acetone as nucleophile and afforded the axially chiral ether **3aa** without obvious reduction of enantioselectivity.

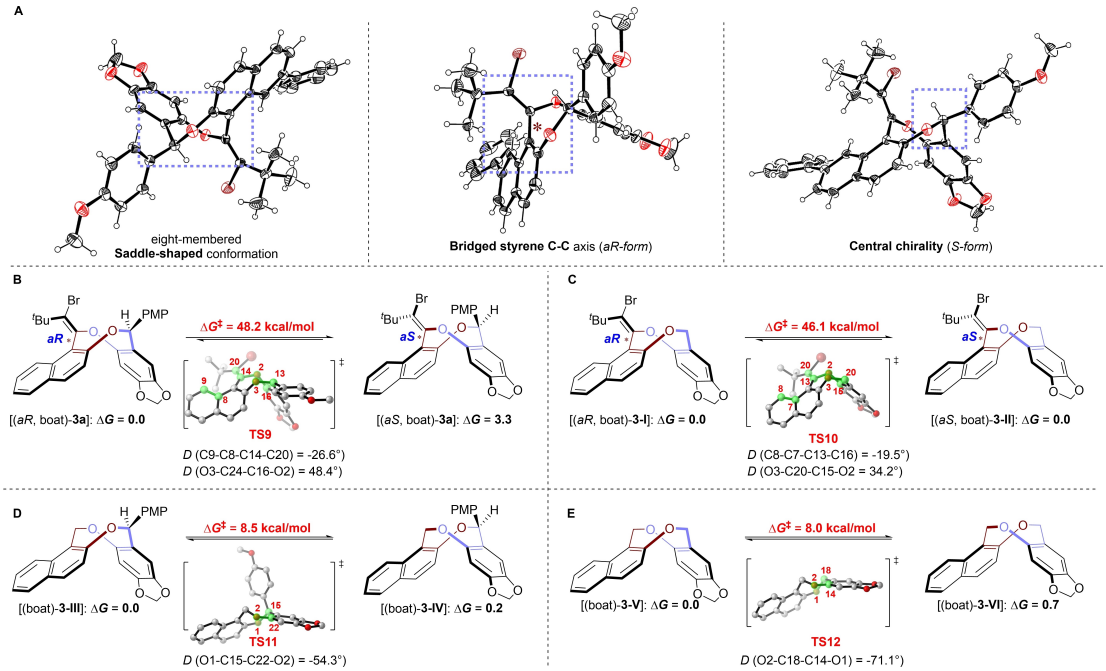


Figure 3. A) X-ray structure of **3o** in different views, thermal ellipsoid at 30% probability level. B–E) DFT calculation of stereochemical stability of the saddle-shaped chiral eight-membered O-heterocycles. All energies were calculated at the M06-2X/6-311+G(d,p)/SMD(toluene)//B3LYP/6-31G(d)/SMD(toluene) level of theory. Values are given in kcal/mol. *D* = dihedral angle.

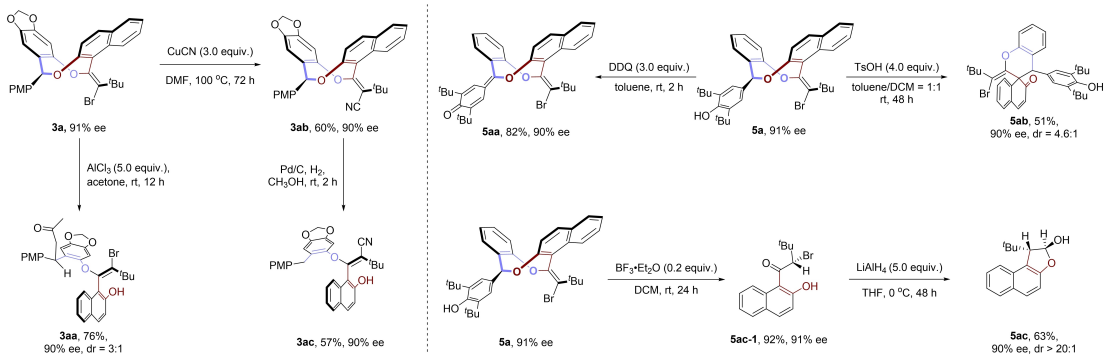


Figure 4. Synthetic transformations.

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The cyanation of **3a** was conducted with CuCN in DMF at 100 °C to produce **3ab** in moderate yield with similar enantioselectivity. The sequential hydrogenation of **3ab** gave access to the axially chiral ether **3ac**. Moreover, product **5a** could be selectively oxidized to furnish the corresponding *p*-quinone methide **5aa** in excellent yields with similar enantioselectivity. The simple treatment of **5a** with *p*-toluenesulfonic acid delivered spirocyclic **5ab**. It was assumed to be realized through an acid-promoted remote elimination and subsequent addition to the in situ generated *p*-quinone methide by the naphthol moiety in an enolate C-alkylation fashion. When a stronger Lewis acid such as $BF_3{\cdot}OEt_2$ was used, compound **5a** was reverted back to the starting VQM intermediate. This intermediate was then trapped by H_2O, leading to the formation of **5ac–1** after undergoing keto-enol tautomerism. Subsequently, the chiral dihydronaphthofuran **5ac**, featuring vicinal tertiary carbon stereocenters, was obtained through a reaction with lithium aluminum hydride. The structures of **3aa**, **5ab**, and **5ac** were unambiguously determined by single-crystal X-ray diffraction analysis.[108]

Conclusions

In conclusion, we illustrated a novel asymmetric synthetic approach of inherently chiral eight-membered O-heterocycles through cross-[4+4] cycloaddition of quinone methides with vinylidene *ortho*-quinone methides. The developed reactions proceeded in high efficiency with excellent selectivity. As a novel oxa-analogue of tetraphenylene, the desired eight-membered O-heterocycles also featured the saddle-shaped orientation and exhibited high geometric rigidity. The theoretical investigation revealed that the hindered rotation of the embedded axially chiral styrene helped to fix the conformation of the target eight-membered O-heterocycles. This study offered a general synthesis approach to the inherently chiral medium-sized ring system and expanded the application scope of QMs for higher-order cycloaddition reactions.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: asymmetric synthesis · atroposelectivity · eight-membered rings · inherent chirality · organocatalysis

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